

Sky & Space: Class 7

The second year for classes that did Sky & Space 6: light, the Sun as an engine, tides and craters, and the year the class earns a telescope — ending at the eyepiece.

10

SESSIONS

36

CONCEPTS TAUGHT

11 h 20 min

TOTAL

2 h 40 min

OUTDOORS

School: Middle (6–8)

AUDIENCE

Moon: start when the moon is **18.75–22.5 days old** (🌕 waning gibbous → 🌔 last quarter)

ASSUMED ALREADY KNOWN

Measuring the Earth

Zero Shadow Day

Equinoxes & Solstices

Solar Eclipses

Aryabhata's Astronomy

Indian Star & Constellation Names

India in Space

How We See

Maps & Scale Models

Closed under prerequisites: 34 concepts in total are taken as prior knowledge.

1 Light is a traveller

CLASSROOM · 1 H 20 MIN · DAY 5

KIT Projector · Paper, pencils, protractor

L2 Speed, Distance & Time

Speed = distance ÷ time: know two, find the third. Thunder after lightning gives the storm's distance — the same trick, scaled up, turns light's travel time into cosmic distances.

L1 Rainbows & Colours

White sunlight is a mixture of all colours — raindrops or a prism fan it out into a rainbow. Colour is a property of the light itself, not paint added to it.

L2 Light & Lookback Time

Light is fast but finite — 8 minutes from the Sun, 4 years from the nearest star. Every telescope is a time machine: we see objects as they were, not as they are.

L3 Reflection & Refraction

Mirrors bounce light by a fixed rule; glass bends it as it slows down. These two behaviours are the entire toolkit from which every telescope is built.

TRAINER NOTE Open with the eight minutes: the sunshine on their desk left the Sun before the period started. Speed-distance-time is maths class made real — have them compute the Moon's 1.3 seconds themselves before you reveal it. Then a mirror, a glass of water with a pencil in it, and a prism at the window: bounce, bend, split. Do not name lenses or telescopes yet, but know that this entire year rests on this one period — every instrument they will touch is these three tricks in a tube.

2 The Sun is an engine

CLASSROOM · 40 MIN · DAY 12

KIT Projector · Paper, pencils, protractor

L1 The Sun Powers Earth

Sunlight drives nearly everything here: plants, winds, rain, and the warmth of the day. Grasping the Sun as Earth's engine is the first reason to care where it is in the sky.

L3 The Inverse-Square Law

Double the distance, quarter the light — brightness falls as $1/d^2$. This single rule connects a star's true power, its distance, and how bright it looks to us.

TRAINER NOTE Every joule in the room — the food, the fan, the child — is rerouted sunshine; trace one chain (sun → grass → roti → cricket shot) on the board and let them build two more. Then inverse-square with a torch and a grid drawn on card: double the distance, a quarter the brightness, measured not preached. Land the punchline on the solar system they already know: this is WHY Venus bakes and Neptune freezes, and they can now say by how much.

3 Glass: lenses, mirrors, and the telescope

INDOOR LAB · 1 H 20 MIN · DAY 26

KIT Binoculars · Dobsonian reflector · Eyepiece set · Paper, pencils, protractor

L2 Binocular Astronomy

A 7×50 binocular — 7× magnification, 50 mm objectives — shows Moon craters, Jupiter's moons and hundreds of clusters. The best first instrument, and many observers' favourite forever.

L3 Lenses, Mirrors & Focal Length

A curved lens or mirror gathers parallel starlight to a focus; the focal length sets image scale. Real images on a screen — this is where 'optics' becomes 'instrument'.

L2 First Light: Using a Telescope

Align the finder in daylight, start with the lowest power, centre on something bright, focus slowly. Ten minutes of drill separates a magical first night from a frustrating one.

L3 Refractors & Reflectors

Refractors use a lens, reflectors a mirror, catadioptrics both. Each design trades cost, size, and image quality differently — there is no single 'best telescope'.

TRAINER NOTE A magnifier focusing the window onto a sheet of paper — a real, upside-down, full-colour image of the world on a page — is the moment optics stops being a diagram. Build from it: two lenses make a refractor, a curved mirror makes their Dobsonian. Then hands on the actual instruments, in daylight, at a distant building: finding, focusing, and the discipline of not touching glass. Every child moves the Dobsonian and focuses the binoculars TODAY; the sky night is not the place to learn where the focuser is. Say out loud that magnification is the least important number — they will meet the 500x marketing lie at home.

4 Our star, up close and safely

SOLAR SESSION · 1 H 20 MIN · DAY 33

SKY needs **Any moon** · The Sun

KIT Dobsonian reflector · Solar filter (front aperture) · Paper, pencils, protractor

L2 The Sun is a Star

Our Sun is an ordinary star seen close-up — a million-km fusion furnace. Every night-sky star is a sun; many have their own planets.

L3 Sunspots & Solar Activity

The Sun's magnetic weather: spots, flares and mass ejections rising and falling on an 11-year cycle. It drives aurorae and can rattle satellites and power grids.

L3 Solar Observing (Safely)

NEVER look at the Sun without certified filters — projection and proper solar film show sunspots safely; hydrogen-alpha scopes reveal prominences. Safety protocol is the first lesson.

TRAINER NOTE The Sun is a star; the stars are suns — say it once at the start and once at the end, because it reorders the whole sky in their heads. Projection first: the whole class around one projected disc, hunting sunspots, before anyone queues at the filtered eyepiece. Safety is theatre and must be theatrical: inspect the front-aperture filter in front of them, cap the finder in front of them, and tell them about the eyepiece-filter cracks of old. Sunspots bigger than the Earth, counted by children — and if the disc is blank that day, teach them that the quiet Sun is a real observation too. Log the count; the aurora session will want it.

5 Lights the atmosphere makes

CLASSROOM · 40 MIN · DAY 47

KIT Projector · Paper, pencils, protractor

L1 Weather vs the Sky

Clouds, rain and haze live a few kilometres up; the Moon, planets and stars lie far beyond. Weather hides the sky but never touches it — astronomy begins where weather ends.

L2 The Ocean of Air

We live at the bottom of a thin skin of air that thins with height and fades into space around 100 km up. It burns meteors, blurs stars, bends light — and keeps us alive.

L2 Halos & Coronae

A big 22° ring around the Moon or Sun is a halo from ice crystals; a small rainbow-fringed disc hugging the Moon is a corona from water droplets. Free atmospheric optics, often before rain.

L3 Aurorae

Solar-wind particles funnelled down Earth's magnetic field make the upper air glow in curtains of green and red. Strong storms have pushed aurorae as far south as Ladakh.

TRAINER NOTE The sorting lesson: clouds, rain and rainbows live in the bottom few kilometres; the Moon and stars live behind all of it — most children file 'sky' as one drawer and this period gives them two. Halos are the payoff: a ring round the Moon is ice crystals in OUR air, not something the Moon does, and once told, children start reporting them within the fortnight — put a halo-report corner on the classroom wall. Close with aurora as the two drawers connecting: the Sun's tantrums (their own sunspot log) lighting the top of the air. Be honest that they will likely never see one from India; honest scarcity beats false promise.

6 The Moon pulls, the Moon remembers

INDOOR LAB · 1 H 20 MIN · DAY 68

KIT Projector · Paper, pencils, protractor · Measuring tape

L2 Craters & Maria

The Moon's dark 'seas' are ancient lava plains; its craters are asteroid scars, preserved for billions of years with no air or water to erase them. The face of the Moon is a history book.

L3 Tides

The Moon pulls Earth's near side harder than its far side, raising two water bulges — hence two tides a day. Sun-Moon alignment makes spring tides; right angles make neaps.

L2 Asteroids & the Kuiper Belt

Rocky leftovers between Mars and Jupiter, icy ones beyond Neptune (Pluto's realm). Small-body surveys are one field where amateurs still make discoveries.

L3 Impacts on Earth

Earth gets hit too — Lonar Lake in Maharashtra is a 50,000-year-old impact crater, and the dinosaurs met a 10-km asteroid. Erosion, not luck, hides our scars.

TRAINER NOTE Two faces of the same neighbour. First the pull: the Moon's gravity stretches the ocean, and the trap to disarm is 'one bulge' — there are two, and a child holding the diagram will ask why; have the answer ready (the far side is left behind, not pushed). Then the scars: flour tray, cocoa dusting, and marbles dropped from different heights — every crater is round regardless of impact angle, which is the question someone always asks. Their craters against Chandrayaan photographs, then the sting: the Moon keeps every scar for four billion years because it has none of the air and weather we sorted out in the last session. Earth got hit just as hard and healed.

7 Other moons, and the rings

CLASSROOM · 40 MIN · DAY 82

KIT Projector · Paper, pencils, protractor

L3 Galilean Moons

Jupiter's four big moons shuffle visibly from night to night — a miniature solar system in any small telescope, and Galileo's proof that not everything orbits Earth.

L3 Rings of Saturn

A sheet of ice chunks thinner, in proportion, than paper — spanning 280,000 km. The rings tilt through a 29-year cycle and vanish edge-on, as they did for Galileo.

TRAINER NOTE Galileo pointed a telescope worse than the school's binoculars at Jupiter and watched four dots change sides night after night — show his actual notebook sketches and let the class decode them before you explain. The point is seismic: things orbit something other than the Earth, and a class-7 child can re-collect the evidence themselves. Saturn's rings: ice rubble, shepherded, and thinner in proportion than a sheet of paper — a rubble ring around a football-sized Saturn would be tissue-thin. Promise Jupiter's moons at the sky night only if a planet is actually up; check [whatsup.html](#) before you promise.

KIT Star chart / planisphere · Red torch · Paper, pencils, protractor

L2 The Eye as a Detector

Cone cells see colour but need light; rod cells see faint things only in grey. That's why nebulae look ghostly-grey in the eyepiece yet glow red and blue in photographs.

L2 Star Names & Bayer Letters

Bright stars carry ancient proper names (many Arabic: Aldebaran, Deneb); astronomers label stars α , β , γ by brightness within each constellation. Two naming systems, one sky.

L2 Dark Adaptation & Averted Vision

Eyes take 20–30 minutes to reach full night sensitivity, and one white-light glance resets them. Looking slightly beside a faint object uses your more sensitive off-centre retina.

L2 Star Charts & Planetarium Apps

Reading a sky map: orientation, scale, magnitudes, and matching chart to sky. A planisphere or app turns the catalogue sky into 'what's above me right now'.

L2 Observing Logs & Sketching

Date, time, sky conditions, instrument, and what you actually saw — a log turns stray sessions into a record of growth, and sketching trains the eye to see more.

TRAINER NOTE The unglamorous session that decides whether the sky night succeeds. Blackout the lab and spend real minutes in the dark: let them experience faint things surfacing as their eyes adapt, then ruin it with one white torch flash — now the red torch needs no policing, they have felt why. Planisphere drill as a game: set the date and hour, race to name what is up tonight. Star names give the chart its handles; the observing log is the year's closer — rule the first page together and make the sky night its first entry. A log they keep is the difference between an outing and an astronomer.

9 Night at the eyepiece

OBSERVING NIGHT · 1 H 20 MIN · DAY 103

SKY needs **Thin crescent** · The Moon, Milky Way, Open clusters, Constellations

KIT Dobsonian reflector · Binoculars · Eyepiece set · Star chart / planisphere · Red torch · Smartphone · Phone-to-eyepiece adapter · Tripod

L2 The Milky Way Band

From a dark site a faint luminous band arches across the sky — the combined light of billions of stars in our galaxy's disc, seen from inside it.

L2 Earthshine

The ghostly glow filling the dark part of a crescent Moon is sunlight reflected off Earth — 'the old Moon in the new Moon's arms'. You are seeing our planet light another world.

L3 Nebulae & Star Clusters

Glowing gas clouds where stars are born; open clusters of young siblings; ancient globular swarms of hundreds of thousands. The Milky Way's visible ecology.

L2 Lunar Observing

Craters, maria, mountain shadows — the Moon is best along the terminator, where low sunlight throws relief into drama. Full moon is actually the flattest, worst time.

L2 Smartphone Astrophotography

A phone held (or bracketed) at the eyepiece captures the Moon and planets; night modes catch constellations and even the Milky Way. The zero-cost gateway to imaging.

TRAINER NOTE The year's summit, and the crescent is chosen, not luck: a terminator full of shadowed craters early, earthshine to explain while they queue ('the ghost light is Earth shining on the Moon's night'), and it sets in time to leave the sky dark for the faint stuff. Run the Moon at the Dobsonian first and let every child take one phone photo through the eyepiece — the adapter beats hand-holding, and that photo does more for the subject at home than any circular. Then binoculars up the Milky Way band to a bright cluster: teach them it is grainy with stars, a thing no screen conveys. Everyone writes their log entry ON SITE, red torch, before the bus — memory edits, logs do not. Cloud date agreed with the school in advance.

10 Build a sundial, own the week

HANDS-ON / BUILD · 1 H 20 MIN · DAY 124

KIT Gnomon / shadow stick · Paper, pencils, protractor · Measuring tape · Projector

L2 The Navagrahas

The nine 'grahas': Sun, Moon, five visible planets, plus Rahu and Ketu — not nine planets, but the nine movers of the sky as ancient India catalogued them.

L3 Building Sundials

A sundial's gnomon must parallel Earth's axis — tilted at your latitude, aimed at the pole. Get that right and horizontal, vertical or equatorial dials all read true.

L3 Vara & the 7-Day Week

The week comes from the seven classical 'planets' ruling hours in rotation — Ravivar/Sunday to Shanivar/Saturday match across India, Rome and Babylon. Astronomy fossilised in every calendar.

TRAINER NOTE The closing workshop ties this year's optics-and-instruments spirit back to class 6's shadow stick: the gnomon must now be TILTED to the pole star's height — their latitude, which they measured last year — and suddenly one dial-plate works all year. Each pair builds and calibrates a card sundial against the school's true-north line at noon. Then the aha that hides in the calendar: Sunday, Monday, Saturday — the week is the seven moving lights of the navagraha, sitting in their pocket diary all along. Teach Rahu and Ketu as the eclipse points, the old astronomers' bookkeeping for a real geometry. Present it as heritage of observation; the astrology argument gets its own airing in a later year, when they can follow it.

FULL KIT LIST

Binoculars	sessions 3, 9
Dobsonian reflector	sessions 3, 4, 9
Eyepiece set	sessions 3, 9
Gnomon / shadow stick (<i>consumable</i>)	sessions 10
Measuring tape	sessions 6, 10
Smartphone	sessions 9
Phone-to-eyepiece adapter	sessions 9
Projector	sessions 1, 2, 5, 6, 7, 10
Red torch	sessions 8, 9
Solar filter (front aperture)	sessions 4
Star chart / planisphere	sessions 8, 9
Paper, pencils, protractor (<i>consumable</i>)	sessions 1, 2, 3, 4, 5, 6, 7, 8, 10
Tripod	sessions 9
